Course Code	Course Title	L	T	P	C
PRES701L	Research Methodology	3	1	0	4
Pre-requisite	NIL	Syllabus version			
		1.0			

(Common to SCE, SCHEME, SCOPE, SELECT, SENSE, SCORE, SMEC)

Course Objectives:

1. To gain insights into how scientific research is conducted.
2. To help in the critical review of the literature and assessing the research trends, quality and extension potential of research and equip students to undertake research.
3. To learn and understand the basic statistics involved in data presentation.
4. To identify the influencing factors or determinants of research parameters.
5. To test the significance, validity and reliability of the research results.
6. To help in the documentation of research results.

Course Outcomes:

On the completion of this course, the student will be able to:

1. Ability to critically evaluate current research and propose possible alternate directions for further work.
2. Ability to develop hypotheses and methodology for research.
3. Ability to comprehend and deal with complex research issues to communicate their scientific results clearly for peer review.

Module:1	Introduction	3 Hours

Introduction to research; Definitions and characteristics of research; Types of research – Descriptive vs. Analytical, Applied vs. Fundamental, Quantitative vs. Qualitative, Conceptual vs. Empirical, Main components of any research work.

Module:2	Research Formulation	4 Hours

Defining and formulating the research problem: Selecting the problem – Necessity of defining the problem, Literature review: Importance of literature review in defining a problem, Primary and secondary sources – reviews, treatise, and monographs, patents – web as source – searching in the web – Critical literature review – Identifying gap areas from literature review.

Module:3	Research Design	3 Hours

Essentials of Research Design, Need for Research Design, Classifications of Research Design: causations and Experimental Design, Errors in Research Design, Types of Research Errors.

Module:4	Data Collection	7 Hours
Observation and Collection of data, Methods of data collection, Different types of variables, Sampling Methods- Data Processing and Analysis strategies - Data Analysis with Statistical Packages, Hypothesis-testing, Generalization and Interpretation.		
Module:5	**Quantitative Methods**	**8 Hours**
Probability Distributions, Fundamentals of Statistical Analysis and Inference, Multivariate methods, Concepts of Correlation and Regression, Fundamentals of Time Series Analysis and Spectral Analysis.		
Module:6	**Optimization**	**10 Hours**
Introduction to evolutionary algorithms - Fundamentals of Genetic algorithms, Particle Swarm Optimization, Simulated Annealing, Introduction to Neural Networks, Neural Network-based optimization, Introduction to Fuzzy sets and Fuzzy Logic, Optimization of fuzzy logic.		
Module:7	**Ethics**	**3 Hours**
Ethical issues - Copy right - Intellectual property rights and patent law - Reproduction of published material - Plagiarism - Citation and acknowledgement - Reproducibility and accountability.		
Module:8	**Research Report Writing & Latex**	**7 Hours**
Structure and Components of Research Report, Types of Report, Layout of Research Report, Mechanism of writing a research report, Latex Commands, Latex Styles Files, Presentations in Latex, Report writing on Latex.		

Reference Books

1. Ranjit Kumar, “Research Methodology: A Step-by-Step Guide for Beginners”, SAGE Publications Ltd; Fourth edition, 2014.
2. C.R. Kothari, “Research Methodology: Methods and Techniques”, NEW AGE; Second Edition, 2014.
3. Peter Pruzan, “Research Methodology: The Aims, Practices and Ethics of Science”, Springer; First Edition, 2016.
4. Andries P. Engelbrecht “Computational Intelligence: An Introduction”, Wiley-Blackwell, 2nd Edition, 2007.
5. Anthony J. Hayter “Probability and Statistics for Engineers and Scientists”, CENGAGE Learning Custom Publishing; 4th edition, 2011.
6. Bernard C. Beins and Maureen A. McCarthy “Research Methods and Statistics” Pearson, 2011.
7. Colin Neville “The Complete Guide to Referencing and Avoiding Plagiarism”, Open University

Press, 2007. 8. Dinesh Kumar, Research Methods for Successful PhD, River Publishers Series in Innovation and Change in Education – Cross-Cultural Perspective, 2017 (ISBN: 9788793609181). 9. Marc van Dongen, LaTeX and Friends, Springer, Feb. 29, 2012 (ISBN 978-3-642-23815-4).
Tutorial work (1 hour per week)
Training on software packages such as Latex, MATLAB and other domain-specific software. Review Paper writing in Latex.
Mode of Evaluation: CAT / Assignment / Quiz / FAT / Seminar / Tutorials

Approved by Academic Council	No. 73	Date	14-03-2024

Approval			
S.No	**Name of the Member**	**Role**	**Signature**
1	**Dr. DAPHNE LOPEZ** Dean, School of Computer Science Engineering and Information Systems VIT, Vellore - 632014.	Dean	
2	**Dr. KUMAR P** Associate Professor Dept of Information Technology and Engineering Manonmaniam Sundaranar University Abishekapatti, Tirunelveli Tirunelveli-627012	External Member	
3	**Dr. NARENDRAN RAJAGOPALAN** Associate Professor Department of Computer Science and Engineering National Institute of Technology NIT Puducherry, Tiruvettakudy, Karaikal 609 609 Pondicherry-609609	External Member	
4	**Dr. NITHYA.S** Associate Professor Grade 2 School of Computer Science Engineering and Information Systems VIT, Vellore - 632014 Information Systems, VIT, Vellore.	Internal Member	
5	**Dr. RAJKUMAR M** Associate Professor Grade 1, School of Computer Science Engineering and Information Systems, VIT, Vellore.	Guide	

Course Code	Course Title	L	T	P	C
PRMC001L	**Research and Publication Ethics**	**2**	**0**	**0**	**2**
Pre-requisite	**NIL**	**Syllabus version**			
		1.0			

(Common to all Research Scholars)

Course Objectives:

1. Course for awareness about the publication ethics and publication misconducts.
2. To understand ethical dilemmas faced in research and publication
3. To understand Intellectual honesty and research integrity.
4. To learn and understand Best Practices for Publication process
5. To identify the impact factor journals, and research metrics

Course Outcomes:

On the completion of this course the student will be able to:

1. Critically evaluate research Publications and ethical issues.
2. Distinguish publication misconduct and scientific research.
3. Analyze how ethical challenges can be addressed in research and publication
4. Delineate limits of data that are collected and utilized during research and publication
5. To comprehend and deal with complex research metrics and data base.

Module:1	**Philosophy and Ethics**	**3 Hours**

Introduction to Philosophy: Definition, nature and scope, concept, branches, Ethics: Moral Philosophy, nature of moral judgments and reactions.

Module:2	**Scientific Conduct**	**5 Hours**

Ethics with respect to Science and Research, Intellectual honesty and Research integrity Scientific misconducts: Falsification, Fabrication and Plagiarism(FFP), Redundant Publications: Duplicate Publications and overlapping Publication, salami slicing, Selective
Reporting misrepresentation of data.

Module:3	**Publication Ethics**	**7 Hours**

Publication Ethics: Definition, introduction and importance, Best Practices/standards setting initiatives and guidelines : Committee on **Publication Ethics** (COPE), World Association of Medical Editors (WAME) etc., Conflicts of interest, Publication misconduct: definition, concept, problems that lead to unethical behavior and vice versa types, Violation of Publication ethics, authorship and contributorship,
Identification of Publication misconduct, complaints and appeals, Predatory Publishers and Journal

Module:4	**Practice 1: OPEN Access Publishing**	**4 Hours**

1. Open access Publications and initiatives
2. SHERPA/ROMEO online resource and check publisher copy right & Self archiving policies
3. Software tool to identify predatory publications developed by SPPU Journal finder/ Journal suggestion tools Viz. JANE, Elsevier journal finder, Springer, Journal suggester etc.,

Module:5	Practice 2:Publication Misconduct	4 Hours
Group Discussion: Subject specific ethical issues , FFP and authorship, Conflict of interest, Complaints and appeals: example fraud from India and abroad, Software		
tools: Use of Plagiarism software Turnitin, Urkund and other open source software tools		
Module:6	**Practice 3: Databases and Research Metrics**	**7 Hours**
Data Bases: Indexing Database, Citation Data Bases: Webof Science, Scopus etc, Impact factor of Journal as per Journal Citation Report, SNIP, SJR, IPP, Cite Score, Metrics: h-index, g index, i10 index, almetrics		

Reference Books

10. Leefmann, J., & Jungert, M. (2020). Research Ethics and Scientific Integrity in Neuroscience. Handbook of Research Ethics and Scientific Integrity, 1013- 1035.
11. Iphofen, R., & Tolich, M. (Eds.). (2018). The SAGE handbook of qualitative research ethics. Sage..
12. Temple, A. (2019). The Postgraduate's Guide to Research Ethics. United Kingdom: Macmillan International Higher Education.
13. Research and Publications Ethics by Santosh Kumkar Yadav, Ane Publications,2020
14. Kara, H. (2018). Research Ethics in the Real World: Euro-Western and Indigenous Perspectives. United Kingdom: Policy Press.
15. Anil. K. Jain, Ethical issues in scientific publication, Indian J Orthop. 2010 Jul- Sep; 44(3): 235–237.
16. Gupta, S., Kamboj, S. (2020). Research and Publication Ethics. (n.p.): Alexis Press LLC.
17. Bos, J. (2020). Research ethics for students in the social sciences (p. 287). Springer Nature.Cavaliere, P., De Souza, D., Fenton, A. L., Giridharan, B., Gralla, C., Inshakova, N., & Zaharuk, G. (2020). Academic Misconduct and Plagiarism: Case Studies from Universities Around the World. Lexington Books.
18. Bretag, T. (Ed.). (2016). Handbook of academic integrity. Springer Singapore.

19. Dobrick, F. M., Fischer, J., & Hagen, L. M. (2018). Research Ethics in the Digital Age. Wiesbaden: Springer.
20. Manual for research and publication ethics in science and engineering, Korean Federation of Science and Technology Societies, Seoul, Korea,2016
21. Kleinert S & Wager E (2011) Responsible research publication: international standards for editors. Promoting Research Integrity in a Global Environment. Imperial College Press / World Scientific Publishing, Singapore (pp 317-28). (ISBN 978-981-4340-97-7)

Mode of Evaluation: CAT / Assignment / Case Study discussion / FAT / Seminar/Tutorials			
Approved by Academic Council	No. 73	Date	14-03-2024

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3	**Dr. NARENDRAN RAJAGOPALAN** Associate Professor Department of Computer Science and Engineering National Institute of Technology NIT Puducherry, Tiruvettakudy, Karaikal 609 609 Pondicherry-609609	External Member	
4	**Dr. NITHYA.S** Associate Professor Grade 2 School of Computer Science Engineering and Information Systems VIT, Vellore - 632014 Information Systems, VIT, Vellore.	Internal Member	
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Office of Academic Research

Details of the Research Scholar					
Name	**POOJA M**			Register No.	**26PHD0374**
Programme	Doctor of Philosophy	School	**SCORE**	Category	**Internal Full Time**
Topic of Research	**Slack-Free Lagrangian Constraint Handling Against Slack-Based Methods for Reducing Qubit Overhead in Quantum Combinatorial Optimization**				

Details of Special Elective (Self-Study Course/ Guide Course)				
COURSE TITLE:	**QUANTUM MACHINE LEARNING**			
Credit Structure (Common to all Special Elective Courses)	**L**	**T**	**P**	**C**
	0	**3**	**0**	**3**
Module 1: Introduction to Quantum Machine Learning				
Introduction to Machine Learning - Introduction to Quantum Computing - Introduction to Quantum Machine Learning - Classical Machine Learning vs Quantum Machine Learning - Quantum Advantage - Applications of Quantum Machine Learning - Challenges in Quantum Machine Learning				
Module 2: Quantum Computing Foundations for Machine Learning				
Qubits - Quantum States - Superposition - Entanglement - Quantum Measurement - Quantum Gates - Quantum Circuits - Multi-Qubit Systems - Quantum Information Encoding				
Module 3: Quantum Data Encoding and Feature Maps				
Classical Data Encoding - Quantum Data Encoding - Basis Encoding - Amplitude Encoding - Angle Encoding - Feature Maps - Quantum Feature Space - Quantum Kernels - Data Preparation for Quantum Machine Learning				
Module 4: Variational Quantum Machine Learning				
Parameterized Quantum Circuits - Variational Quantum Circuits - Trainable Parameters - Quantum Ansatz - Cost Functions - Classical Optimizers - Hybrid Quantum-Classical Learning - Training Variational Quantum Models				
Module 5: Quantum Machine Learning Algorithms				
Quantum Support Vector Machines - Quantum Kernel Methods - Quantum Neural Networks - Variational Quantum Classifiers - Quantum Clustering - Quantum Principal Component Analysis - Quantum Generative Models				
Module 6: Quantum Neural Networks and Hybrid Models				
Quantum Neural Networks - Quantum Layers - Quantum Perceptron's - Variational Quantum Neural Networks - Hybrid Quantum-Classical Models - Classical Neural Networks with Quantum Circuits - Training Hybrid Models				

- Loss Functions and Optimization.
References
1. **Maria Schuld and Francesco Petruccione**, *Machine Learning with Quantum Computers*, 2nd Edition, Springer, 2021. 2. **Maria Schuld and Francesco Petruccione**, *Supervised Learning with Quantum Computers*, Springer, 2018. 3. **Peter Wittek**, *Quantum Machine Learning: What Quantum Computing Means to Data Mining*, 2nd Edition, Academic Press, 2017. 4. **Michael A. Nielsen and Isaac L. Chuang**, *Quantum Computation and Quantum Information*, 10th Anniversary Edition, Cambridge University Press, 2010. 5. **John Preskill**, *Quantum Computing: Lecture Notes*, California Institute of Technology. 6. **Vijay P. Singh**, *Quantum Machine Learning: An Introduction*, Springer.
Mode of Evaluation: CAT / Assignment / Quiz / Seminar / Tutorial /FAT

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Office of Academic Research

Details of the Research Scholar					
Name	**POOJA M**			Register No.	**26PHD0374**
Programme	Doctor of Philosophy	School	**SCORE**	Category	**Internal Full Time**
Topic of Research	**Slack-Free Lagrangian Constraint Handling Against Slack-Based Methods for Reducing Qubit Overhead in Quantum Combinatorial Optimization**				

Details of Special Elective (Self-Study Course/ Guide Course)					
COURSE TITLE:	**QUANTUM OPTIMIZATION**				
Credit Structure (Common to all Special Elective Courses)		**L**	**T**	**P**	**C**
		0	**3**	**0**	**3**
Module 1: Fundamentals of Optimization					
Introduction to Optimization - Objective Functions - Decision Variables - Constraints - Feasible and Infeasible Solutions - Constrained and Unconstrained Optimization - Binary Optimization - Combinatorial Optimization - NP-Hard Problems					
Module 2: Quantum Computing Fundamentals for Optimization					
Introduction to Quantum Computing - Qubits - Quantum States - Superposition - Entanglement - Quantum Gates - Quantum Circuits - Measurement - Quantum Computing for Optimization					
Module 3: QUBO and Ising Model Formulations					
Binary Decision Variables - Quadratic Unconstrained Binary Optimization (QUBO) - QUBO Objective Function - QUBO Matrix - QUBO Formulation - Ising Model - Spin Variables - QUBO to Ising Transformation - Energy Minimization.					
Module 4: Quantum Optimization Algorithms					
Introduction to Quantum Optimization Algorithms - Variational Quantum Algorithms - Quantum Approximate Optimization Algorithm (QAOA) - Variational Quantum Eigen solver (VQE) - CVaR-VQE - Cost Hamiltonian - Mixer Hamiltonian - Parameterized Quantum Circuits - Classical Optimization of Parameters					
Module 5: Constraint Handling in Quantum Optimization					
Types of Constraints - Equality Constraints - Inequality Constraints - Constraint Encoding - Penalty Methods - Penalty Coefficients - Slack Variables - Binary Encoding of Slack Variables - Qubit Overhead - Feasibility and Optimality					

Module 6: Lagrangian Relaxation and Slack-Free Optimization
Lagrangian Relaxation - Lagrange Multipliers - Lagrangian Function - Dual Problem - Lagrangian Duality - Constraint Relaxation - Sub gradient Method - Dual Averaging - Bundle Method - Cutting Plane Method - Augmented Lagrangian - Slack-Free Formulations - Unbalanced Penalty Methods
References
1. **Daniel J. Egger, Jakub Mareček, and Stefan Woerner**, *Warm-Starting Quantum Optimization*, Springer, 2021. 2. **Amin K. R. et al.**, *Quantum Optimization: Theory and Applications*, Springer. 3. **Michael A. Nielsen and Isaac L. Chuang**, *Quantum Computation and Quantum Information*, 10th Anniversary Edition, Cambridge University Press, 2010. 4. **Alexandre Montanaro**, *Quantum Algorithms: An Overview*, Cambridge University Press. 5. **Frank G. S. L. Brandão and Krysta M. Svore**, *Quantum Algorithms for Linear Algebra and Optimization*, Springer. 6. **Peter Wittek**, *Quantum Machine Learning: What Quantum Computing Means to Data Mining*, Academic Press, 2017.
Mode of Evaluation: CAT / Assignment / Quiz / Seminar / Tutorial /FAT

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5	**Dr. RAJKUMAR M** Associate Professor Grade 1, School of Computer Science Engineering and Information Systems, VIT, Vellore.	Guide	

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Office of Academic Research

Details of the Research Scholar					
Name	**POOJA M**			Register No.	26PHD0374
Program	Doctor of Philosophy	School	**SCORE**	Program	**Internal Full Time**
Topic of Research	**Slack-Free Lagrangian Constraint Handling Against Slack-Based Methods for Reducing Qubit Overhead in Quantum Combinatorial Optimization**				

Details of Online Course						
COURSE TITLE:	**Foundations of Deep Learning: Concepts and Applications**					
CODE (If Any):	**Credit (Common to all Online Courses)**	**L**	**T**	**P**	**C**	
		2	**0**	**0**	**2**	
Name of the Instructor	Prof. Ashwini Kodipalli, Prof. Sriram Ganapathy					
Name of the Institution	**IISc Bangalore RV University**					
Name of the Online Platform	NPTEL					
Online Platform Web Link	nptel.ac.in/courses/106108840					
Duration of the Course (Minimum 8 Weeks for NPTEL Platform / 30 Hours for other Online Platforms with Assessment)	12 Weeks					
Start Date	13-12-2026	**End Date**	13-03-2027			
About Instructor						

Prof. Sriram Ganapathy is an Associate Professor in the Department of Electrical Engineering at the Indian Institute of Science (IISc), Bengaluru, and the principal investigator of the LEAP Lab (Learning & Extraction of Acoustic Patterns). He holds a B.Tech. in Electronics and Communication Engineering from the College of Engineering, Trivandrum (2004), an M.E. in Signal Processing from IISc (2006), and a Ph.D. in Electrical and Computer Engineering from Johns Hopkins University (2012). Before joining IISc in 2016, he worked as a Research Staff Member at the IBM Watson Research Center (2011–2015) and served as a Visiting Research Scientist at Google Research India/DeepMind (2022–2024). His research focuses on speech and audio signal processing, deep learning, representation learning, auditory neuroscience, and explainable AI, with applications in speech recognition, speaker recognition, emotion analysis, and acoustic sensing. He has led impactful projects such as modulation filter learning for robust speech models and the Coswara COVID-19 vocal screening initiative. Dr. Ganapathy has authored over 120 peer-reviewed publications, guided award-winning student projects including the Qualcomm Innovation Fellowship, and teaches courses like Speech Information Processing and Machine Learning for Signal Processing. His contributions have been recognized with several honors, including the Verisk AI Faculty Research Award (2021, 2022), the DAE Young Scientist Award (2018), the IEEE SigPort Chief Editorship, and the title of IBM Master Inventor. He is a Senior Member of IEEE and an active member of the International Speech Communication Association (ISCA). Prof. Ashwini Kodipalli

Dr. Ashwini Kodipalli holds a Ph.D. from the Indian Institute of Science (IISc), Bangalore, and is currently a faculty member at RV University. With over 15 years of teaching experience, she is deeply passionate about simplifying complex concepts and making them accessible to students. Her research expertise lies in Biomedical Image Analysis, particularly in the application of advanced Deep Learning algorithms for healthcare.
Course Outline
1. Deep Learning is a core pillar of modern Artificial Intelligence, powering applications in computer vision, natural language processing, healthcare, and robotics. 2. With the rapid expansion of AI-related programs in AICTE-affiliated institutions, it's essential for students to build a strong foundation in this field. 3. This course is designed to guide learners' step-by-step—from the basics of neural networks to advanced architectures like CNNs, RNNs, and Autoencoders. 4. Emphasis is placed on both conceptual clarity and practical implementation using Python and Google Collab. 5. By the end of the course, students will be able to apply deep learning algorithms on real-world data to develop intelligent solutions.
COURSE PLAN:
Week 1- Introduction to Artificial Neural Networks **Week 2-** Optimizers and Regularization Techniques **Week 3-** Convolutional Neural Networks **Week 4-** Variants of CNN Architectures **Week 5-** Explainable AI: Concepts and Techniques **Week 6-** CNN based Segmentation & Object Detection Algorithms **Week 7-** Sequence-to-Sequence models (Recurrent Neural Networks) **Week 8-** Long-short term memory (LSTM) **Week 9-** Natural Language Processing **Week 10**- Autoencoders (AEs) and Variational Autoencoders (VAEs) **Week 11-** Large Language Models (LLMs) **Week 12-** Large Language Models (continued) and Diffusion Models

Approval			
S. No	**Name of the Member**	**Role**	**Signature**
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3	**Dr. NARENDRAN RAJAGOPALAN** Associate Professor Department of Computer	External Member	

	Science and Engineering National Institute of Technology NIT Puducherry, Tiruvettakudy, Karaikal 609 609 Pondicherry-609609		
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